

Supplementary Online Content

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This supplementary material has been provided by the authors to give readers additional information about their work.

eTable 1. Full list of inclusion and exclusion criteria

Inclusion criteria

- Age 40 to 80 years (both males and females; premenopausal women must have a negative pregnancy test before inclusion into the trial, postmenopausal women are defined as women who have not had menstrual bleeding for at least 12 months or have been surgically sterilized)
- Planned screening or diagnostic colonoscopy
- Able to speak and understand German sufficiently to be able to give written informed consent and comply with the trial requirements

Exclusion criteria

- Factors potentially influencing the primary endpoint
 - Diseases/symptoms
 - Chronic inflammatory bowel disease (e.g. Crohn's disease, ulcerative colitis)
 - Colonoscopy due to positive fecal occult blood test
 - History of colorectal cancer
 - Angiodysplasia of the colon
 - Anamnestic or observed blood loss per anum
 - Use of any of the following drugs
 - Within 2 weeks before the trial
 - Anticoagulants (including, but not limited to heparin, vitamin K antagonists [e.g. phenprocoumon, warfarin], direct thrombin inhibitors [e.g. dabigatran], or factor Xa inhibitors [e.g. apixaban, rivaroxaban])
 - Antiplatelet drugs (e.g. clopidogrel, prasugrel, ticlopidine)
 - Within 1 week before the trial
 - Acetylsalicylic acid
 - Within 3 days before the trial
 - NSAIDs including selective COX-2 inhibitors
- Factors potentially affecting the safety
 - Any current clinically relevant signs and symptoms, including
 - Signs and symptoms suggesting acute peptic ulcer disease
 - Known clinically relevant thrombocytopenia
 - Acute infection
 - Volume deficit (dehydration)
 - Any currently present allergy with dermal reactions, pruritus, or urticaria
 - Severe or insufficiently controlled asthma
 - Severe kidney or liver diseases (e.g. GFR < 30 ml/min, liver cirrhosis)
 - Severe, not sufficiently treated heart failure (as judged by the investigator)
 - Severe, poorly controlled arterial hypertension
 - Any other unclear symptoms needing further investigation in the opinion of the investigator
 - Any of the following anamnestic findings
 - History of severe gastrointestinal bleeding

- Known hemorrhagic diathesis, including, but not limited to, hypoprothrombinemia, severe thrombocytopenia, hemophilia
 - Asthma, except for patients who have used acetylsalicylic acid in the past without negative effects
 - Hypersensitivity against salicylic acid or other ingredients of the trial drugs
 - Previous intolerance to NSAIDs including selective COX-2 inhibitors, or antirheumatic medications
 - Severe gout (e.g. recurrent attacks)
 - Hereditary oxaluria
 - Known glucose-6-phosphate dehydrogenase or glutathione peroxidase deficiency
 - Known epilepsy with generalized seizures
 - Severe cardiac diseases (including, but not limited to, myocardial infarction in the past 6 months)
- Intention to use any of the following drugs during the trial
 - Anticoagulants
 - Antiplatelet drugs
 - NSAIDs including selective COX-2 inhibitors
 - Methotrexate ≥ 15 mg/week
 - Systemically administered glucocorticoids
 - Selective serotonin reuptake inhibitors (SSRIs)
 - Valproic acid
- Planned surgery/dental treatment during participation in the trial
- Other factors
 - Known or suspected relevant alcohol abuse
 - Known or suspected illicit drug abuse
 - Pregnancy
 - Suspected non-compliance with the trial procedures
 - Participation in another clinical trial

eTable 2. Numbers of valid FIT results according to day of fecal sampling in relation to day of tablet intake

Group	Fecal sampling, return and testing				Actual day of fecal sampling in relation to day of tablet intake							
	Test	Scheduled Day ^a	Returned N	Valid Result ^b N	Missing N	<0 N	0 N	1 N	2 ^c N	3 N	4 N	>4 N
Intervention N=1075	Quantitative	≤0	1044	1026	82	152	781	9	1	0	0	1
		2	1045	1030	77	2	11	126	758	48	5	3
		3	989	975	77	1	0	9	105	700	65	18
		4	955	944	64	1	0	0	7	93	677	102
		Any	4033	3975	300	156	792	144	871	841	747	124
	Qualitative	≤0	1037	970	78	149	733	8	1	0	0	1
		2	1040	974	75	2	9	121	719	42	4	2
		3	979	922	71	1	0	9	97	663	65	16
		4	955	887	60	1	0	0	7	88	638	93
		Any	4011	3753	284	153	742	138	824	793	707	112
Control N=1059	Quantitative	≤0	1016	1000	80	159	753	5	2	0	0	1
		2	1024	1012	78	2	4	136	747	38	4	3
		3	987	979	75	1	1	5	114	706	58	19
		4	959	947	72	1	0	1	1	104	656	112
		Any	3986	3938	305	163	758	147	864	848	718	135
	Qualitative	≤0	1026	968	76	149	735	5	2	0	0	1
		2	1024	969	72	2	4	126	720	39	3	3
		3	979	934	73	1	1	5	108	677	51	18
		4	956	905	66	1	0	1	1	103	627	106
		Any	3985	3776	287	153	740	137	831	819	681	128

^a Day refers to scheduled day of stool sampling in relation to day of tablet intake

^b Test result satisfying all the following conditions: (i) Valid test result; (ii) Sample collected before colonoscopy (or information not available)

^c Samples used for per-protocol analysis

eTable 3. Sensitivity (Se) and specificity (Sp) for detecting advanced neoplasms (colorectal cancer or advanced adenoma) after single 300 mg aspirin dose or placebo, based on at least one positive test result from up to three stool samples scheduled on day 2, 3 and 4 after intake of aspirin or placebo

Sex	Test	Cutoff [µg Hb/g]	Aspirin group						Placebo group						Difference			
			TP N	FN N	TN N	FP N	Se %	Sp %	TP N	FN N	TN N	FP N	Se %	Sp %	Sensitivity [95% CI] %		Specificity [95% CI] %	
All	Quantitative	10.2	63	50	685	261	55.8	72.4	49	58	774	161	45.8	82.8	10.0	[-3.3, 22.8]	-10.4	[-14.1, -6.6]
	Quantitative	17.0	44	69	816	130	38.9	86.3	35	72	852	83	32.7	91.1	6.2	[-6.4, 18.7]	-4.9	[-7.7, -2.0]
	Qualitative	10.2	53	58	714	222	47.7	76.3	35	73	779	146	32.4	84.2	15.3	[2.3, 27.8]	-7.9	[-11.5, -4.3]
Men	Quantitative	10.2	46	31	311	148	59.7	67.8	28	30	362	102	48.3	78.0	11.5	[-5.5, 27.8]	-10.3	[-15.9, -4.5]
	Quantitative	17.0	34	43	384	75	44.2	83.7	22	36	408	56	37.9	87.9	6.2	[-10.5, 22.4]	-4.3	[-8.8, 0.3]
	Qualitative	10.2	41	34	343	111	54.7	75.6	19	40	372	89	32.2	80.7	22.5	[5.6, 38.0]	-5.1	[-10.5, 0.2]
Women	Quantitative	10.2	17	19	374	113	47.2	76.8	21	28	412	59	42.9	87.5	4.4	[-16.7, 25.1]	-10.7	[-15.4, -5.8]
	Quantitative	17.0	10	26	432	55	27.8	88.7	13	36	444	27	26.5	94.3	1.2	[-17.4, 20.4]	-5.6	[-9.1, -2.0]
	Qualitative	10.2	12	24	371	111	33.3	77.0	16	33	407	57	32.7	87.7	0.7	[-19.0, 20.7]	-10.7	[-15.5, -5.9]

Abbreviations: CI, confidence interval; FN, false negatives; FP, false positives; g, gram; Hb, hemoglobin; Se, sensitivity; Sp, specificity; TN, true negatives; TP, true positives

eTable 4. Sensitivity (Se) and specificity (Sp) for detecting advanced neoplasms (colorectal cancer or advanced adenoma) at day 2 after single 300 mg aspirin dose or placebo. Results of the per-protocol analysis.

Sex	Test	Cutoff [µg Hb/g]	Aspirin group						Placebo group						Difference			
			TP N	FN N	TN N	FP N	Se %	Sp %	TP N	FN N	TN N	FP N	Se %	Sp %	Sensitivity [95% CI] %		Specificity [95% CI] %	
All	Quantitative	10.2	42	56	635	136	42.9	82.4	25	61	691	85	29.1	89.0	13.8	[-0.1, 27.1]	-6.7	[-10.1, -3.2]
	Quantitative	17.0	29	69	704	67	29.6	91.3	18	68	735	41	20.9	94.7	8.7	[-4.0, 20.8]	-3.4	[-5.9, -0.8]
	Qualitative	10.2	30	56	617	119	34.9	83.8	19	65	683	62	22.6	91.7	12.3	[-1.4, 25.4]	-7.8	[-11.2, -4.5]
Men	Quantitative	10.2	31	34	310	78	47.7	79.9	11	34	344	58	24.4	85.6	23.2	[5.0, 39.5]	-5.7	[-10.9, -0.4]
	Quantitative	17.0	23	42	350	38	35.4	90.2	9	36	373	29	20.0	92.8	15.4	[-1.8, 30.9]	-2.6	[-6.5, 1.4]
	Qualitative	10.2	24	33	311	63	42.1	83.2	9	37	353	40	19.6	89.8	22.5	[4.5, 38.6]	-6.7	[-11.5, -1.8]
Women	Quantitative	10.2	11	22	325	58	33.3	84.9	14	27	347	27	34.1	92.8	-0.8	[-21.8, 20.6]	-7.9	[-12.3, -3.4]
	Quantitative	17.0	6	27	354	29	18.2	92.4	9	32	362	12	22.0	96.8	-3.8	[-21.6, 15.0]	-4.4	[-7.6, -1.1]
	Qualitative	10.2	6	23	306	56	20.7	84.5	10	28	330	22	26.3	93.8	-5.6	[-25.1, 15.3]	-9.2	[-13.7, -4.6]

Abbreviations: CI, confidence interval; FN, false negatives; FP, false positives; g, gram; Hb, hemoglobin; Se, sensitivity; Sp, specificity; TN, true negatives; TP, true positives

eTable 5. Positive predictive value (PPV) and negative predictive value (NPV) in % for detecting advanced neoplasms (colorectal cancer or advanced adenoma) at day 2 after single 300 mg aspirin dose or placebo. Results of the per-protocol analysis.

Sex	Test	Cutoff [μg]	Aspirin		Placebo		Difference in PPV [95% CI]		Difference in NPV [95% CI]	
			PPV	NPV	PPV	NPV				
All	Quantitative	10.2	23.6	91.9	22.7	91.9	0.9	[-9.3, 10.7]	0.0	[-2.8, 2.8]
	Quantitative	17.0	30.2	91.1	30.5	91.5	-0.3	[-15.3, 14.2]	-0.5	[-3.3, 2.3]
	Qualitative	10.2	20.1	91.7	23.5	91.3	-3.3	[-14.8, 7.7]	0.4	[-2.6, 3.3]
Men	Quantitative	10.2	28.4	90.1	15.9	91.0	12.5	[-0.2, 24.1]	-0.9	[-5.2, 3.4]
	Quantitative	17.0	37.7	89.3	23.7	91.2	14.0	[-4.9, 31.1]	-1.9	[-6.1, 2.2]
	Qualitative	10.2	27.6	90.4	18.4	90.5	9.2	[-5.9, 22.8]	-0.1	[-4.4, 4.2]
Women	Quantitative	10.2	15.9	93.7	34.1	92.8	-	[-34.7, -1.3]	0.9	[-2.9, 4.6]
	Quantitative	17.0	17.1	92.9	42.9	91.9	-	[-48.4, -0.7]	1.0	[-2.7, 4.8]
	Qualitative	10.2	9.7	93.0	31.2	92.2	-	[-38.9, -3.9]	0.8	[-3.2, 4.8]

Abbreviations: CI, confidence interval; FIT, fecal immunochemical test; g, gram; Hb, hemoglobin; NPV, negative predictive value; PPV, positive predictive value

eTable 6. Sensitivity analysis for the effect of study site. Estimated generalized linear mixed model random intercept standard deviations (St. Dev.) and corresponding p-values from the likelihood ratio test are shown. The null hypothesis of standard deviation equal to zero is tested against the alternative hypothesis that the standard deviation is greater than zero. Agreement between test and colonoscopy result for the detection of advanced neoplasms (colorectal cancer or advanced adenoma) at day 2 after single 300 mg aspirin dose or placebo are considered successful outcomes. Results of the intention-to-screen analysis.

Sex	Test	Cutoff [μg Hb/g]	Sensitivity		Specificity	
			St. Dev.	p-value	St. Dev.	p-value
All	Quantitative	10.2	0.00E+00	0.50	5.49E-03	0.38
	Quantitative	17.0	1.97E-01	0.16	0.00E+00	0.50
	Qualitative	10.2	0.00E+00	0.50	9.23E-03	0.30
Men	Quantitative	10.2	0.00E+00	0.50	1.00E-14	0.50
	Quantitative	17.0	0.00E+00	0.50	1.47E-02	0.40
	Qualitative	10.2	0.00E+00	0.50	1.13E-02	0.35
Women	Quantitative	10.2	0.00E+00	0.50	1.79E-12	0.50
	Quantitative	17.0	1.67E-01	0.40	0.00E+00	0.50
	Qualitative	10.2	4.35E-16	0.50	0.00E+00	0.50